Poisson model $\qquad N \neq n$

$$T = \{t_1, \dots, t_n\}$$

$$\text{data} := \begin{cases} T \geq 1 \Rightarrow \#\text{ of years} \\ T[N] > 0 \Rightarrow \#\text{ events in } T_i \text{ year} \end{cases}$$

$N_t \sim Pois(\lambda) \Rightarrow$ # of floods per year is modeled after a poisson dist.

$$P(N_t = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \lambda = E(N_t) \Rightarrow \text{expected \# floods per } t \text{ year.}$$

likelihood: $P(N_1, \dots, N_t \mid \lambda) = \prod_{t=1}^{T} \frac{\lambda^{N_t} e^{-\lambda}}{N_t!}$

$$\cancel{\log(\lambda)} \Leftrightarrow \underbrace{\lambda^{\sum N_t}}_{\lambda^{\text{total events}}} \cdot \underbrace{e^{-T\lambda}}_{e^{\#\text{ years}\cdot\lambda}} \cdot \prod_{t=1}^{T} \frac{1}{N_t!}$$

$\lambda \sim Gamma(2, 1) \qquad E(\lambda) = 2 = V(\lambda)$

I believe there are 2 flood events per year

posterior

$$\lambda \mid N \sim Gamma(2 + \sum N_t, 1 + T)$$

in stan $\log(\text{prior}) + \log(\text{likelihood}) = \log(\text{posterior})$